UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCE – IBRI

MATHEMATICS FOR COMPUTING/MATHEMATIS FOR IT

MATH2101/MATH201. QUIZ – 2

Max Marks 7.5
Max Time: 30 mins

Name:_____ **ID No:**_____

Choose the correct answer from the given. **[5x0.5=2.5Marks]**

1. If $nC2 = nC4$, then the value of $nC6 =$ $2+4=6$ $6C6=1$
   a) 7 b) 1 c) 21 d) 7^2

2. For a student travelling from Ibri to Muscat there are 9 different buses. The number of ways the student can travel from Ibri to Muscat and back from Muscat to Ibri, if he must take the same bus.
   a) 9 b) 10 c) 19 d) 90 $I \xrightarrow{9} M$

3. The number of ways the letters of the word *BARKA* can be arranged is. $\frac{5!}{2!}$
   a) 5! b) $\frac{5!}{2!}$ c) 5! × 2! d) 5! + 2!

4. Suppose that a committee is to be constituted from 6 men and 4 women. How many ways are there to form a committee with 6 members if it must have equal number of men and women?
   a) $6C3 \times 4C3$ b) $6C2 \times 4C2$ c) $10C4$ d) $10C3$ 3w 3m $4C3 \times 6C3$

5. If $\frac{1}{8!} + \frac{1}{9!} = \frac{x}{10!}$, then $x =$ $\frac{1}{8!} + \frac{1}{9\times 8!} = \frac{x}{10\times 9\times 8!}$ $1+\frac{1}{9} = \frac{x}{90}$ $\frac{10}{9} = \frac{x}{90}$ $\frac{9x}{9} = \frac{900}{9}$ $x = 100$
   a) 90 b) 81 c) 100 d) 72

$2\frac{1}{2}$

Solve the questions with necessary steps.

6. There are 28 different time periods during which classes at a university can be scheduled. If there are 539 different classes, how many different rooms will be needed? **[1.5 Marks]**

$\left[\frac{539-1}{28}\right]+1$

$[19.21]+1$

$19+1$

20

$1\frac{1}{2}$

7. How many bit strings of length 6 contain. [1.5 Marks]

a. exactly five 1s?

b. at least five 1s?

$\frac{n!}{r!(n-r)!}$

(a) $\frac{6!}{5!\,1!} = 6$

$\frac{6!}{5!(6-5)!}$

$\boxed{\frac{6!}{5!\,1!}}$

(b) $\frac{6!}{5!\,1!} + \frac{6!}{6!\,0!}$

$6 + 1$

7

$\frac{1}{2}$

8. Find the number of arrangements of the letters of the word INCOGNITO. [2 Marks]

a. How many of these start with G?

b. How many of these have vowels together?

c. How many of these have vowels not together?

Incognito

Total $= \frac{9!}{2!\,2!\,2!} = 45360$

(a) (G) Inconito

$\cancel{\frac{9!}{2!\,2!\,2!} \times 1}$

$\cancel{45360 \times 1}$

$1 \times \frac{8!}{2!\,2!\,2!} = 5040$

(b) (Ioio) ncgot

$\frac{6!}{2!} \times \frac{4!}{2!\,2!}$

$= 2160$

(c) Total − Vowels together

$45360 - 2160$

$= 43200$

2

-o0o-